

Kafka – DevOps

TRAINING & CONSULTING

COMPREHENSIVE IT CONSULTING SERVICES

CORE EXPERTISE CLOUD



AI



EA



NOSQL



Streaming
Platform



Big Data

HENRY POTSANGBAM

IT ARCHITECT & CORPORATE TRAINER | 25+ YEARS EXPERIENCE

AI, Enterprise Architecture, NOSQL, Streaming Platform & Big Data

ACCREDITED CERTIFICATIONS



TOGAF - Certified



Confluent
Accredited - Kafka



Ververica
Apache FLink



Certified Cassandra
Administrator Associate



MapR Hadoop
Cluster Administrator



IBM Certified
Application Developer



IBM Certified
Solution Developer

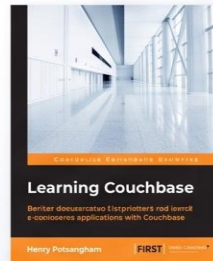


SAP Certified Netweaver
Development Consultant



CIPM - Certificate in
Project Management

AUTHORSHIP & NEW CERTIFICATION



Learning Couchbase
by Henry Potsangbam.

Design document and implement real world
e-commerce applications with Couchbase



GEN AI 360
CERTIFICATION:
LANGCHAIN &
VECTOR DATABASES
IN PRODUCTION



Clientele

Tos



THOMSON REUTERS



GE Healthcare



Apache Kafka Training — 32 Hours

Total Duration **32**
Hours

1. Foundations 6h

Messaging systems, PubSub vs P2P, Kafka architecture, brokers, topics, partitions, replication & CLI.

 Single Node & CLI Exercise

2. Cluster Setup 2h

Zookeeper cluster concepts and multi-node Kafka deployment across AWS EC2 instances.

 Multi-Broker Setup

3. Producers & Consumers 4h

Java API development, sync/async producers, custom partitioners, serializers, and rebalancing.

 Java API Coding

4. Config & Tuning 4h

Compacted topics, compression, broker/producer tuning parameters, log directories, and DR.



5. Security & Ops 4h

Administration tools, Kafka Manager, troubleshooting, ACL authorization, and SSL wire encryption.

 ACL & SSL Labs

6. Kafka Connect 4h

Data pipelines architecture, File, JDBC, and Elasticsearch connectors integration.

 JDBC & Elastic Connect

7. Stream Processing 4h

Kafka Streams architecture, KStream, KTable, ksqlDB servers, queries, and stream-table joins.

 ksqlDB & Streams

8. Schema Governance 4h

Schema Registry, Avro schemas, compatibility rules, subjects, and Confluent Control Center.

 Avro & Registry Lab

Introduce Yourself.

Tos

Name

Year of Experience.

Skills Level

Java / Linux / docker

Messaging System / Kafka

Expectation, if any.

Note: Basic knowledge of Java & Linux are required.

Schedule

Tos

Time	
9.00 – 11.00 AM	Session I
11.00 AM to 11.15 AM	Tea Break
11.15 AM to 1.00 PM	Session II



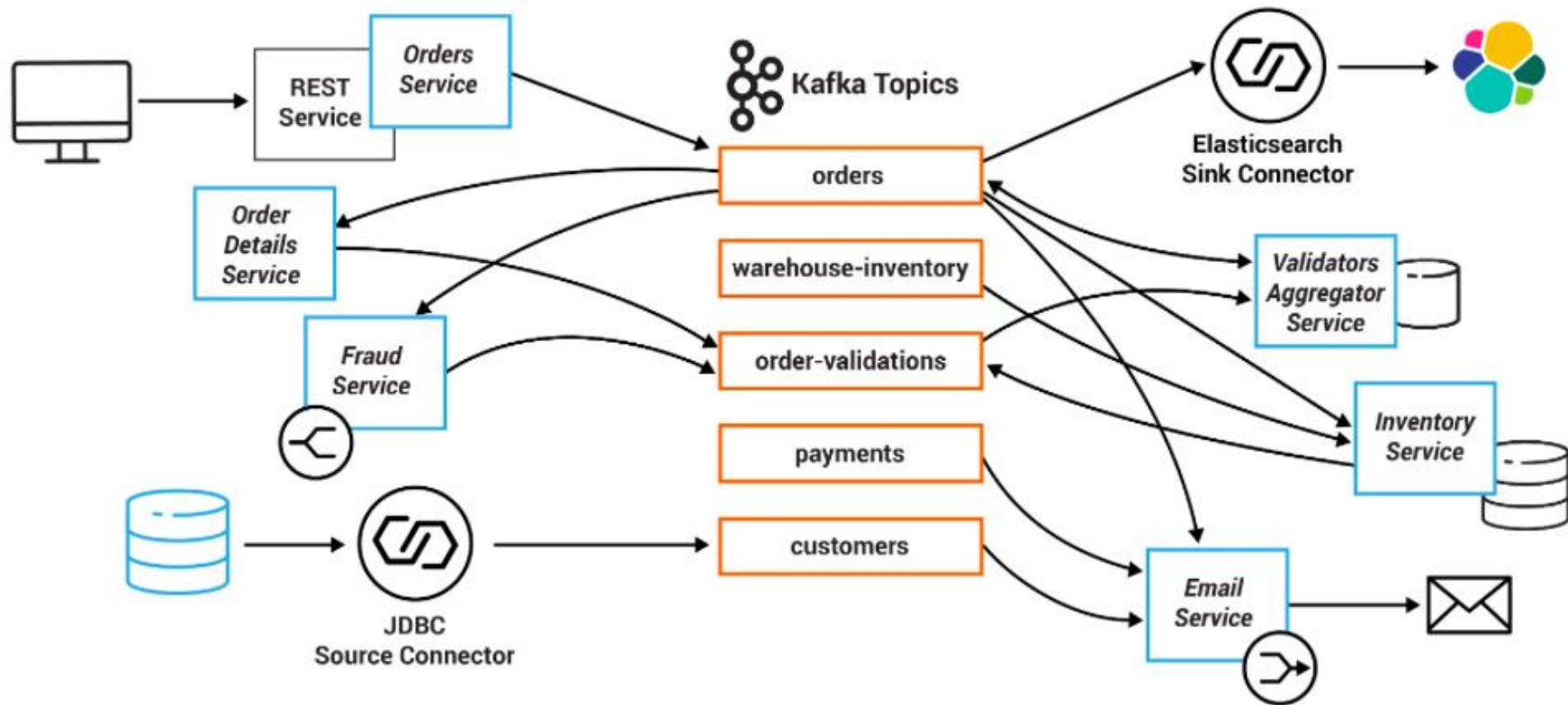
20 September 2026

Kafka – An Overview

- ❖ 1/3 of all Fortune 500 companies
- ❖ Top ten travel companies, 7 of ten top banks, 8 of ten top insurance companies, 9 of ten top telecom companies
- ❖ LinkedIn, Microsoft and Netflix process 4 comma message a day with Kafka (1,000,000,000,000)
- ❖ Real-time streams of data, used to collect big data or to do real time analysis (or both)

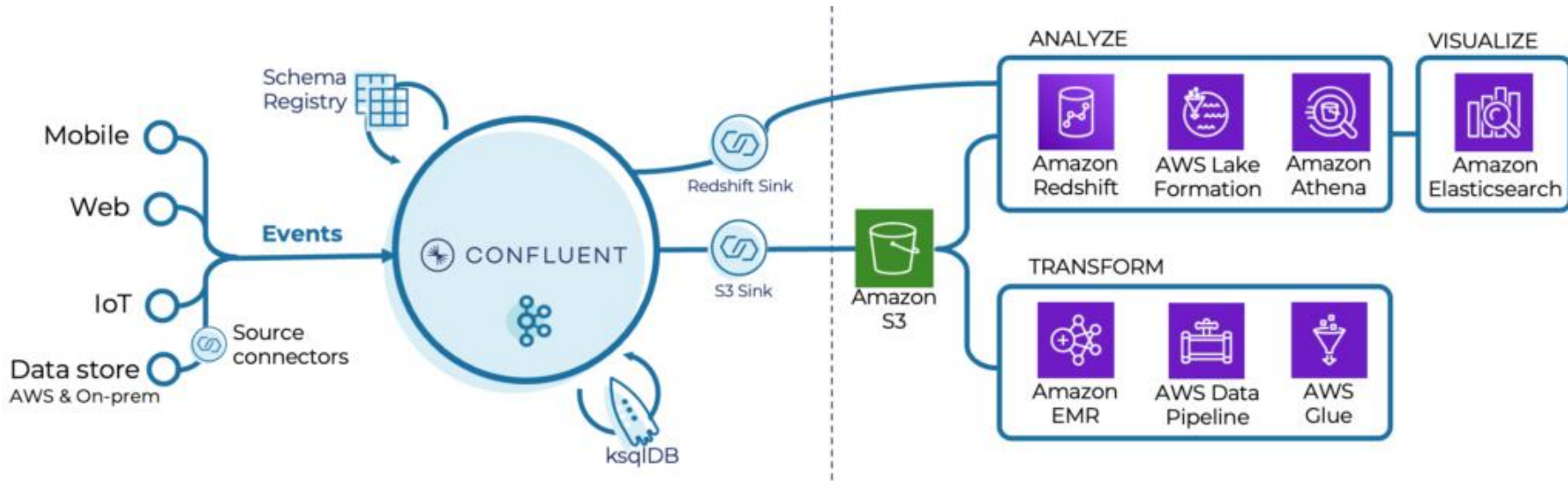
Why Kafka is Needed? - MicroService

Tos



Why Kafka is Needed? – Big Data

Tos



Why Kafka is Needed?

- ❖ Apache Kafka is a fast, scalable, durable, and fault- tolerant publish-subscribe messaging system
- ❖ Real time streaming data processed for real time analytics
- ❖ Service calls, track every call, IOT sensors
- ❖ Kafka is often used instead of JMS, RabbitMQ and AMQP
- ❖ higher throughput, reliability and replication

Why is Kafka needed? 2

- ❖ Kafka can work in combination with
 - Flume/Flafka, Spark Streaming, Storm, HBase and Spark for real-time analysis and processing of streaming data
 - Feed your data lakes with data streams
- ❖ Kafka brokers support massive message streams for follow-up analysis in Hadoop or Spark
- ❖ Kafka Streaming (subproject) can be used for real-time analytics

- ❖ Build real-time streaming applications that react to streams
 - ❖ Real-time data analytics
 - ❖ Transform, react, aggregate, join real-time data flows
 - ❖ Feed events to CEP for complex event processing
 - ❖ Feed data lakes
- ❖ Build real-time streaming data pipe-lines
 - ❖ Enable in-memory micro services (actors, [Akka](#), Vert.x, Qbit, RxJava)

Why is Kafka Popular?

- ❖ ***Great performance***
- ❖ Operational Simplicity, easy to setup and use, easy to reason
- ❖ Stable, Reliable Durability,
- ❖ Flexible Publish-subscribe/queue (scales with N-number of consumer groups),
- ❖ Robust Replication,
- ❖ Works well with systems that have data streams to process, aggregate, transform & load into other stores

Why is Kafka so fast?

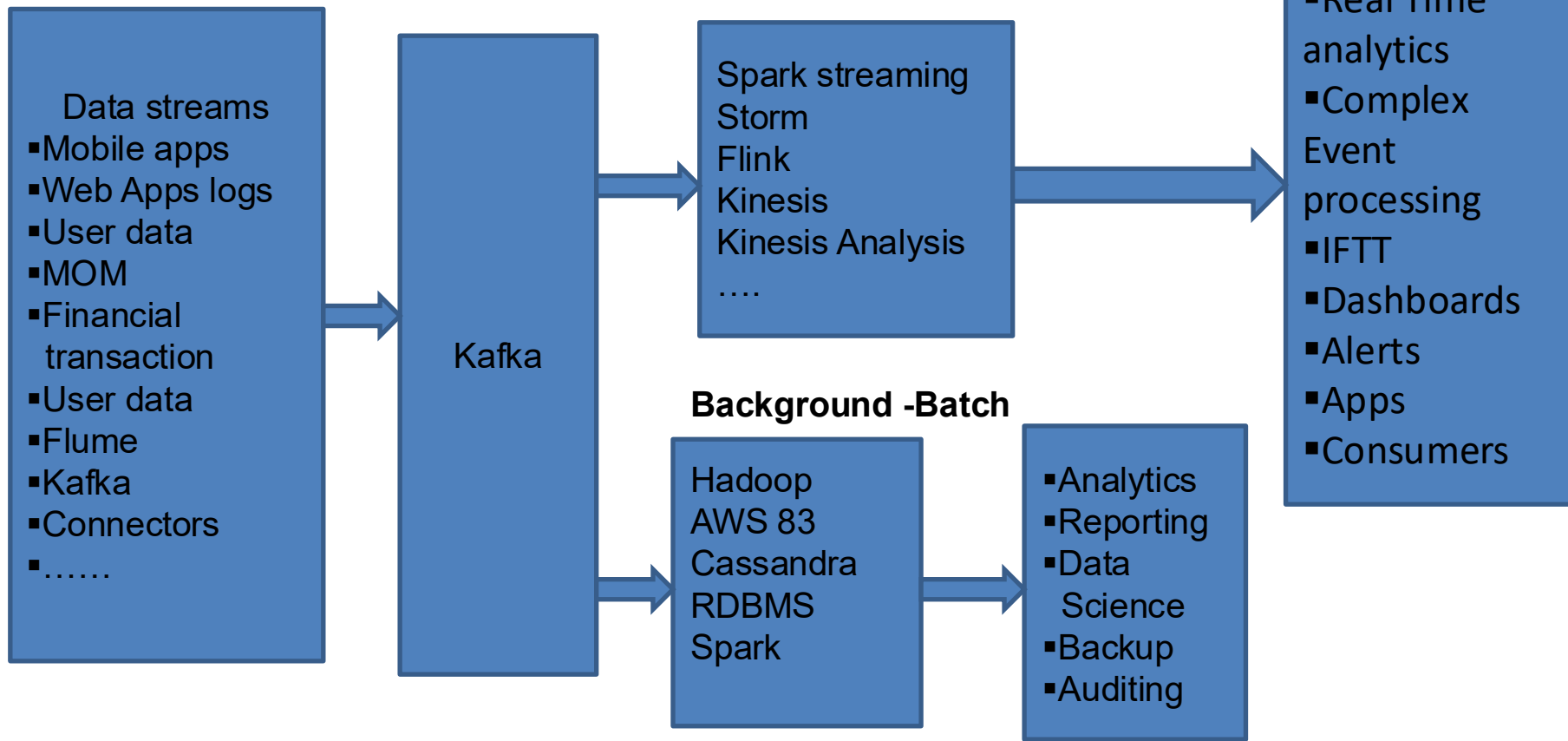
- ❖ **Zero Copy** - calls the OS kernel direct rather to move data fast
- ❖ **Batch Data in Chunks** - Batches data into chunks
 - ❖ end to end from Producer to file system to Consumer
 - ❖ Provides More efficient data compression. Reduces I/O latency
- ❖ **Sequential Disk Writes** - Avoids Random Disk Access
 - ❖ writes to immutable commit log. No slow disk seeking. No random I/O operations. Disk accessed in sequential manner
- ❖ **Horizontal Scale** - uses 100s to thousands of partitions for a single topic
 - ❖ spread out to thousands of servers
 - ❖ handle massive load

Kafka: A Stream Data Platform

Kafka Streaming Architecture

Tos

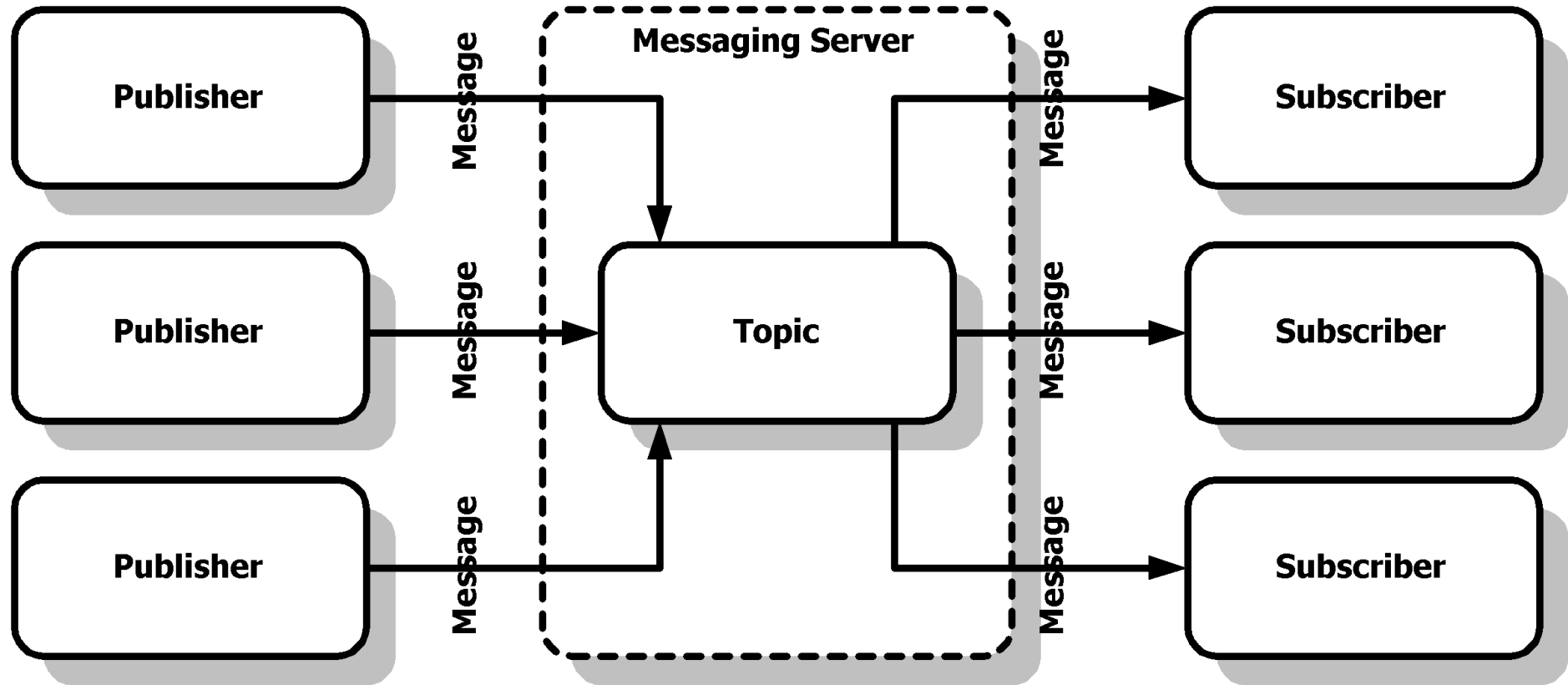
Fast Lane-Real Time-operational



What is Kafka?

What is Kafka?

Tos



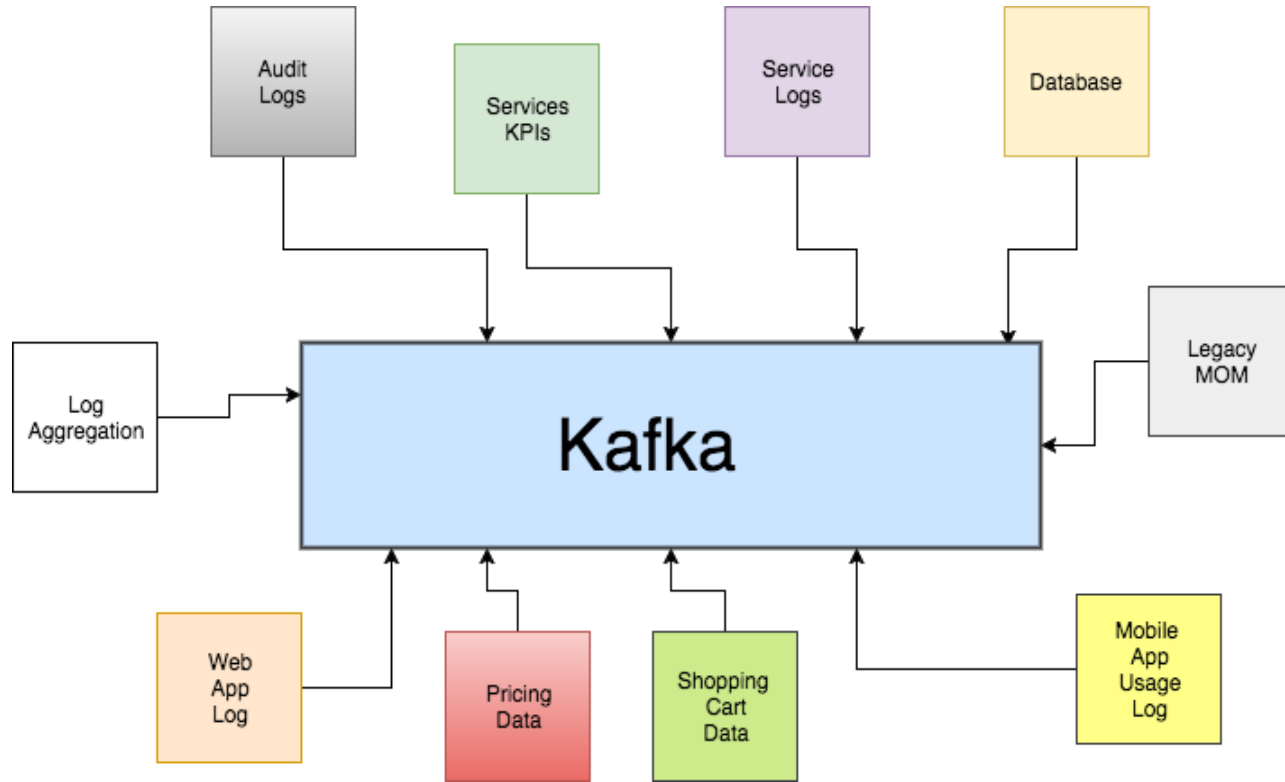
- ❖ Distributed Streaming Platform
 - ❖ Publish and Subscribe to streams of records
 - ❖ Fault tolerant storage
 - ❖ Replicates Topic Log Partitions to multiple servers
 - ❖ Process records as they occur
 - ❖ Fast, efficient IO, batching, compression, and more
- ❖ Used to decouple data streams

- ❖ Kafka decouple data streams
- ❖ producers don't know about consumers
- ❖ Flexible message consumption
 - ❖ Kafka broker delegates log partition offset (location) to Consumers (clients)

- ❖ Feeding of high-latency daily or hourly data analysis into Spark, Hadoop, etc.
- ❖ Feeding micro services real-time messages
- ❖ Sending events to CEP system
- ❖ Feeding data to do real-time analytic systems
- ❖ Up to date dashboards and summaries
- ❖ At same time

Kafka Decoupling Data Streams

Tos



Don't couple the stream

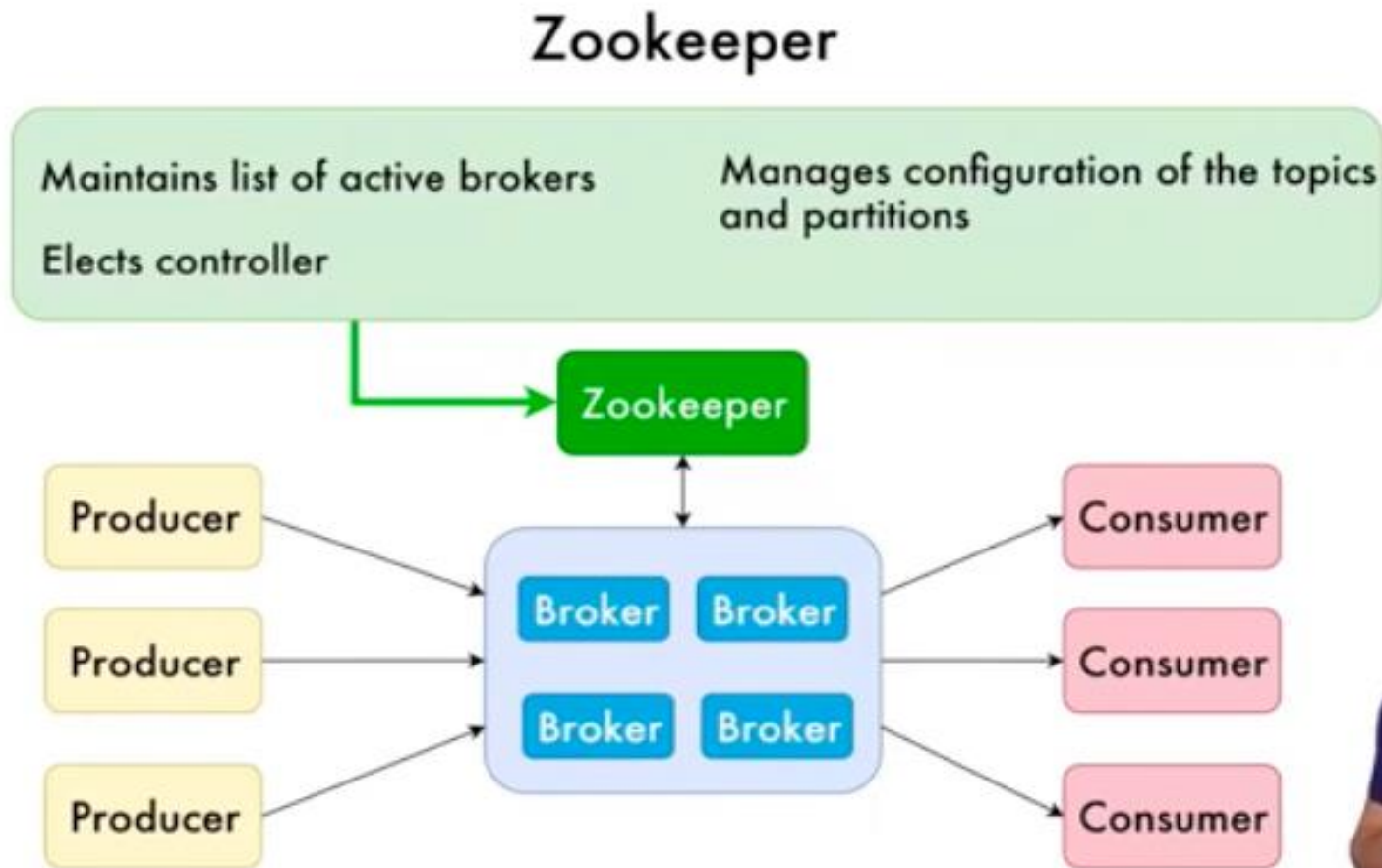


- ❖ Kafka communication from clients and servers wire protocol over TCP protocol
- ❖ Protocol versioned
- ❖ Maintains backwards compatibility
- ❖ Many languages supported
- ❖ Kafka REST proxy allows easy integration (not part of core)
- ❖ Also provides Avro/Schema registry support via Kafka ecosystem (not part of core)

Kafka Architecture

Kafka: Topics, Producers, and Consumers

Tos



- ❖ **Broker:** Kafka server that runs in a Kafka Cluster. Brokers form a cluster. Cluster consists on many Kafka Brokers on many servers.
- ❖ **Zoo Keeper:** Does coordination of brokers/cluster topology. Consistent file system for configuration information and leadership election for Broker Topic Partition Leaders

Installation steps

Installing Java

```
#tar -xvf jdk-8u45-linux-x64.tar.gz -C /opt
```

Set in the path variable and JAVA_HOME

Include in the profile a follow

```
[root@tos opt]# more ~/.bashrc
# .bashrc

# User specific aliases and functions

alias rm='rm -i'
alias cp='cp -i'
alias mv='mv -i'

export JAVA_HOME=/opt/jdk1.8.0_45

export PATH=$PATH:$JAVA_HOME/bin
# Source global definitions
if [ -f /etc/bashrc ]; then
    . /etc/bashrc
fi
[root@tos opt]#
```

Installing Zookeeper

The following example installs Zookeeper with a basic configuration in `/opt/zookeeper`, storing its data in `/opt/data/zookeeper`:

```
# tar -xvf zookeeper-3.4.12.tar.gz -C /opt
```

```
# mv zookeeper-3.4.12 /opt/zookeeper
```

```
# mkdir -p /opt/data/zookeeper
```

```
# vi /opt/zookeeper/conf/zoo.cfg
```

```
tickTime=2000
```

```
dataDir=/opt/data/zookeeper
```

```
clientPort=2181
```

```
# /opt/zookeeper/bin/zkServer.sh start
```

```
[root@tos opt]# /opt/zookeeper/bin/zkServer.sh start
ZooKeeper JMX enabled by default
Using config: /opt/zookeeper/bin/../conf/zoo.cfg
Starting zookeeper ... STARTED
[root@tos opt]#
```

Installation steps..

Installing a Kafka Broker

The following example installs Kafka in `/opt/kafka`, configured to use the

Zookeeper server started previously and to store the message log segments stored in `/tmp/kafka-logs`:

```
# tar -zxf kafka_2.12-1.1.0.tgz -C /opt
```

```
# mv kafka_2.12-1.1.0 /opt/kafka
```

```
# mkdir /opt/data/kafka-logs
```

```
# /opt/kafka/bin/kafka-server-start.sh -daemon /opt/kafka/config/server.properties
```

```
[root@tos opt]# /opt/kafka/bin/kafka-server-start.sh -daemon /opt/kafka/config/s
erver.properties
[root@tos opt]# jps
3476 Kafka
3499 Jps
2895 QuorumPeerMain
[root@tos opt]#
```

Lab : Installation of Kafka